25901396 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACMAT200 - BUSINESS MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. Which diagram is obtained when class frequencies are plotted against midpoints of the corresponding class intervals for a frequency distribution? [K] / [CO1]

2. Find the class marks for the class intervals $21-25$ and $26-30$. [U] / [CO1]

3. Find the slope of the straight line that passes through the points $(2,2)$ and $(3,7)$. [U] / [CO2]

4. Check whether the point $(1,8)$ lies on the line $y=2x+6$. [U] / [CO2]

5. What is an iso-cost line in a Linear Programming Problem? [K] / [CO3]

6. If A is a 3×1 matrix and B is a 1×3 matrix, find the order of the product AB. [K] / [CO4]

7. Check whether the martrix $A=\begin{bmatrix}3 & 1\\ 6 & 2\end{bmatrix}$ is singular or not. [U] / [CO4]

8. Express the supply and demand schedules in matrix form: [U] / [CO4]

Supply: $-6\,p + 24\,q = 213$

Demand: $15\,p + 5\,q = 525$

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

9. A poll of 36 students from one class was conducted to determine how they commute to school. Draw a Bar Graph to represent the given information. [U] / [CO1]

Mode	Cycle	Car	Walking	School Bus
No. of Students	6	4	10	16

10. The monthly production of scooters by a company in the last five months of a particular year is as given below. Represent the data by a Line Chart. [U] / [CO1]

August– 575, September– 425, October– 400, November– 700, December– 625.

11. Solve the simultaneous equations: [U] / [CO2]

$$x + y = 12$$
$$x - y = 2$$

12. Henry has an income of ₹ 36 which he can spend on two goods labelled A and B with prices ₹ 2 and ₹6 respectively. Identify the budget line and sketch the budget set. [U] / [CO3]

13. Determine the matrix $2A - B$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ -2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 5 & 7 \end{bmatrix}$. [U] / [CO4]

14. Solve the following system of linear equations : [U] / [CO4]

$$2\,x_1 + 4\,x_2 = 10$$
$$3\,x_1 + 5\,x_2 = 12$$

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

15. Construct a pie chart to represent the following data showing the percentage distribution of a company's expenditure. [U] / [CO1]
    Raw materials – $40\,\%$
    Wages – $25\,\%$
    Transport – $20\,\%$
    Marketing – $5\,\%$
    Miscellaneous – $10\,\%$

16. A manufacturer makes two products A and B. Product A requires 5 hours for processing, 3 hours for assembling and 4 hours for packaging. Product B requires 2 hours for processing, 12 hours for assembling and 8 hours for packaging. The plant has an availability of 40 hours for processing, 60 hours for assembling and 48 hours for packaging. The profit margin for Product A and Product B is ₹ 7 and ₹ 21 respectively. Express the data as a Linear Programming Problem to determine the number of units of each product to be produced so as to maximize the profit. [A] / [CO3]

17. Find the inverse of the matrix $A = \begin{bmatrix} 2 & 7 & 8 \\ 6 & 3 & 5 \\ 1 & 4 & 9 \end{bmatrix}$. [U] / [CO4]

18. If $A = \begin{bmatrix} 14 & 8 & -3 \\ 6 & 19 & 11 \\ 20 & -1 & 18 \end{bmatrix}$ and $B = \begin{bmatrix} 9 & 4 & 5 \\ -8 & 16 & 7 \\ 13 & 2 & 12 \end{bmatrix}$, find A^T and B^T. Also, verify that $(A^T)^T = A$ and $(B^T)^T = B$. [U] / [CO4]

[10x2 = 20]